

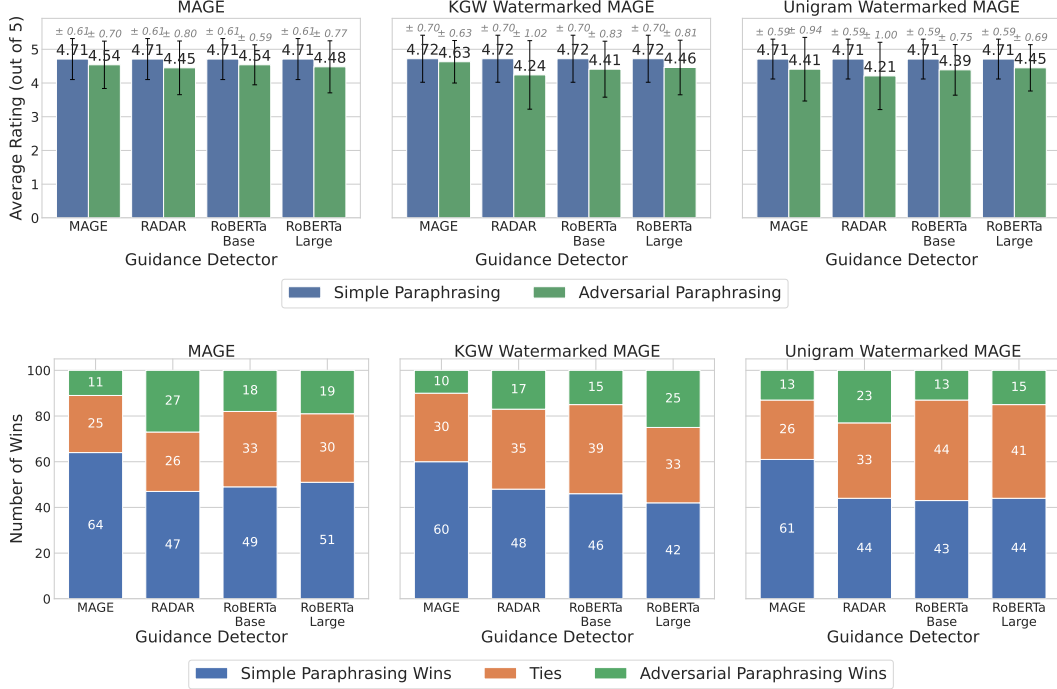


Figure 5: **GPT-4o automated text quality evaluations comparing simple and adversarial paraphrases.** The top row shows Likert-scale ratings for overall quality and semantic similarity to the original text. Though a slight trade off in text quality can be seen, the error bars show that the difference is not statistically significant. The bottom row presents head-to-head win rates, where in most cases, simple paraphrases outperform adversarial paraphrases less than half of the times.

Auto-Rating with GPT-4o. In order to simulate the human perception of text quality (i.e., alignment with the “gold standard” of fluency and coherence), we employ GPT-4o [23], as a judge LLM for automatic quality evaluations [32, 5, 13] with custom system and user prompts (see Appendix G). The judge model is tasked to rate the paraphrases when compared to their original corresponding AI text on a Likert scale of 1-5, in terms of quality and semantic similarity. The first row of Figure 5 shows the quality ratings for both baseline simple paraphrasing and adversarial paraphrasing. Though we observe a slight tradeoff in the text quality when compared to simple paraphrasing, in 87% of the times—averaged across all three datasets and four guidance detectors—adversarial paraphrases were rated 4 or 5 out of 5 (see Appendix H for detailed rating for each guidance detector and dataset). Note that the error bar in the figure shows that this difference between simple and adversarial paraphrasing is not statistically significant. While adversarial paraphrasing can lead to a higher perplexity score when compared to simple paraphrasing, our auto-rater study shows that both the paraphrases have a comparable text quality, making our attack a practical one.

Win Rate Analysis. To further compare the quality of simple and adversarial paraphrases, we conduct pair-wise evaluations using GPT-4o as a judge to compute their win rates [9]. Each pair consists of a simple and an adversarial paraphrase of an AI text, where the judge assigns a win, lose, or tie for the paraphrases. As shown in the second row of Figure 5, simple paraphrases win only less than half of the time in most cases. This finding reinforces the conclusion that adversarial paraphrasing can effectively evade detection with a slight tradeoff in text quality when compared to the prior simple paraphrasing baseline.

6 Conclusion

With our comprehensive experiments, we demonstrate that our proposed **Adversarial Paraphrasing** is a universally transferable and effective attack for humanizing AI-generated text. Our text quality study shows that adversarial paraphrasing can drastically reduce detection rates with slight or no degradation in text quality majority of the time. Our findings underscore the vulnerability of existing detectors in the presence of a strong adversary. In the future, we believe our method can contribute to generating adversarial datasets for improving the robustness of trained detectors.

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